Efficacy of the FIFA cooling break heat policy during an intermittent treadmill football simulation in hot conditions in trained males

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ABSTRACT

Objective To evaluate the efficacy of the Fédération Internationale de Football Association (FIFA) cooling break policy against alternative cooling configurations in attenuating thermal strain during simulated football in the heat.

Methods 12 males (age: 27±6 years, $\dot{V}O_{2peak}$: 61±7 mL/kg/min) completed five 90 min intermittent treadmill football match simulations in 40°C and 41% relative humidity (32°C wet-bulb globe temperature) with different cooling configurations: regular match without cooling breaks (REG), 3 min breaks without cooling ($BRK_{no\text{-}cool}$), 3 min breaks with cooling (BRK_{cool}: current FIFA policy; chilled fluid ingestion and ice towel across neck and shoulders), 5 min extended half-time without cooling breaks ($ExtHT_{only}$) and 3 min cooling breaks with 5 min ExtHT ($ExtHT_{cool}$). Rectal temperature (T_{re}), heart rate, whole-body sweat rate (WBSR) and rating of perceived exertion (RPE) were recorded. Data are presented as mean (95% CIs).

Results Final T_{re} was lower in $BRK_{no\text{-}cool}$ (0.20°C (0.01, 0.39), p=0.038), BRK_{cool} (0.39°C (0.21, 0.57), p<0.001) and $ExtHT_{cool}$ (0.40°C (0.22, 0.58), p<0.001) than REG (39.1°C (38.8, 39.3)). Mean T_{re} was lower in $ExtHT_{cool}$ (38.2°C (38.0, 38.4)) than BRK_{cool} (38.3°C (38.1, 38.5), p=0.018), $BRK_{no\text{-}cool}$ and $ExtHT_{only}$ (38.4°C (38.2, 38.6), p<0.001) and REG (38.5°C (38.3, 38.7), p<0.001). Mean heart rate was lower during BRK_{cool} (6 beats/min (4, 7), p<0.001) and $ExtHT_{cool}$ (7 beats/min (6, 8), p<0.001) compared with REG. WBSR was comparable across trials (p≥0.07) and RPE was attenuated during BRK_{cool} (0.4 (0.1, 0.7), p=0.004) and $ExtHT_{cool}$ (0.5 (0.2, 0.7), p=0.002), compared with REG.

Conclusion BRK_{cool} and $ExtHT_{cool}$ attenuated thermal, cardiovascular and perceptual strain during a simulated football match in the heat. Additional strategies may be required in field settings or under harsher conditions.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Amateur and elite football players often play in hot and/or humid environments that lead to reductions in physical performance and increase the risk of exertional heat illness.

⇒ To reduce the adverse effects of playing football under environmental heat stress, the Fédération Internationale de Football Association (FIFA) Medical Committee implemented brief in-play cooling breaks when the outdoor wet-bulb globe temperature (WBGT) reaches 32°C.

WHAT THIS STUDY ADDS

⇒ This study evaluated the FIFA cooling break policy by implementing different cooling configurations during simulated matches at 32°C WBGT conditions, to determine the most effective approach for attenuating thermal strain in trained males. The current FIFA cooling break policy leads to a reduction in thermal, cardiovascular and perceptual strain, with the combination of the 3 min cooling breaks and a 5 min extension to half-time further reducing thermal and cardiovascular strain.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Findings from this study show that despite the cooling breaks and extended half-time attenuating physiological and perceptual strain during a simulated football match, future research may seek to evaluate different cooling break approaches in real-world settings.

INTRODUCTION

Football matches are often played in hot and/or humid environments during both domestic (eg, A-League, Major League Soccer) and international (eg, Brazil 2014 World Cup and Egypt 2019 African Cup of Nations) competitions. Such environments, combined with the high levels of metabolic heat production during the match, can hinder the physical performance of football players and increase the risk of exertional heat illness.[1–3] The development of exertional heat illness occurs across a spectrum of medical conditions that vary in severity from muscle cramps, to heat exhaustion and in extreme cases, heat stroke, which can be life-threatening.[4] Previous observations of semiprofessional and elite male football players have shown that core temperature can exceed 39.5°C during match play in hot conditions with ambient temperatures from 36°C to 43°C.[5 6] Exertional heat illness has been shown to occur during football training and play, with incidence rates increasing under environmental heat stress.[7 8] As such, the implementation of strategies to improve player safety under heat stress is required,[9 10] which may, in turn, contribute to maintain physical performance or attenuate its decline.

Several strategies for mitigating the adverse effects of environmental heat stress have been investigated in athletes including heat acclimation, precooling and per cooling, and the use of hydration plans.[11–13]

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In elite sport, governing bodies also use extreme heat policies to attenuate the effects of environmental heat stress, such as in-play modifications, schedule changes or even cancellations when particular environmental conditions occur (eg, International Tennis Federation recommends suspension of play when wet-bulb globe temperature (WBGT) exceeds 32.2°C). Similarly, the Fédération Internationale de Football Association (FIFA) has implemented brief in-play cooling breaks as part of their extreme heat policy.[14] These 3 min breaks in the 30th and 75th min of a football match provide the opportunity for both per cooling and fluid intake when the WBGT reaches 32°C. More specifically, the FIFA extreme heat policy states that for each cooling break ice-water-soaked towels, cooler boxes and cold bottled water be made available for both teams and officials.[14]

Chalmers *et al*[9] have previously demonstrated that in comparison to a football match simulated on a treadmill without cooling breaks, the implementation of 3 min breaks used with either chilled fluid ingestion or chilled fluid ingestion and an ice towel draped around the neck and shoulders, attenuated the rise in mean rectal temperature (T_{re}) by −0.25°C and −0.28°C, respectively. The authors also provided evidence for the implementation of an extended half-time (by 5 min) to attenuate the rise of mean T_{re} by −0.21°C, relative to a simulated match without cooling breaks.[9] Extending half-time rather than implementing 3 min breaks during play may be preferred from a match continuity perspective, as it limits disruptions in play and allows for a longer period of cooling in air-conditioned changing rooms. Of note, the previous study evaluated the efficacy of different cooling strategies to attenuate the rise in T_{re} in 30°C WBGT conditions, which are below the current FIFA-specific policy threshold (32°C WBGT) for implementing cooling breaks during major international tournaments. Moreover, the previous study did not compare the effectiveness of a 3 min break with and without cooling, to determine whether the attenuation of thermal strain (ie, core and skin temperature (T_{sk})) stems from a stoppage in play or a stoppage in play with the implementation of cooling strategies.

Given elevated temperatures featured in the European Championships in Germany (2024),[15] and will likely be a focus of the FIFA Men's World Cup in the United States, Canada and Mexico (2026),[16–18] as well as in international play thereafter due to the effects of climate change,[16] it is imperative to evaluate the FIFA cooling break heat policy under the conditions for which it was developed. Therefore, the aim of this study was to evaluate the FIFA cooling break policy using an intermittent treadmill protocol during different match configurations in 32°C WBGT conditions, to determine the most effective approach for attenuating thermal and cardiovascular (ie, heart rate) strain in trained males. It was hypothesised that implementing either the 3 min cooling breaks or extending half-time by 5 min would significantly attenuate the increase in core temperature, T_{sk} and heart rate, compared with a simulated football match without additional breaks or with breaks without cooling ($BRK_{no\text{-}cool}$). It was further hypothesised that the combination of 3 min breaks and a longer half-time would significantly attenuate thermal and cardiovascular strain in comparison to the other trials.

METHODS

Participants

15 trained males volunteered for the study, but 3 withdrew (personal reasons n=2; injury n=1) before completing all experimental trials.[19] As such, data are presented from 12 participants (mean±SD (range): age 27±6 years (21–43), height 1.78±0.07 m (1.66–1.88), mass 74.1±11.7 kg (58.1–91.9), peak oxygen consumption ($\dot{V}O_{2peak}$) 61±7 mL/kg/min (52 – 72), max heart rate 195±8 beats/min (177–208)).

Study design

Following a preliminary trial which consisted of a treadmill test to assess $\dot{V}O_{2peak}$ and 10 min familiarisation to the intermittent treadmill protocol match simulation, participants completed five experimental trials in 40°C±1°C and 41%±1% relative humidity (RH; 32°C WBGT) with 2.0 m/s airflow in a randomised counterbalanced order (figure 1). Excluding the 10 min warm-up (1.0 km), the total distance of each football-specific[20] experimental trial, modified for high-level/elite athletes[21 22] playing in the heat,[5] was 10.7 km, evenly distributed across two 45 min halves (figure 2). All trials were conducted in an environmental chamber and separated by a minimum of 48 hours.

Preliminary trial

On arrival at the laboratory, height and body mass were measured with a wall-mounted stadiometer (Seca 220, Germany) and digital scale (KW Industrial platform scales, Atweigh, Victoria, Australia), respectively. Thereafter, participants were fitted with a heart rate monitor chest strap (HRM Dual, Garmin International, Garmin, Olathe, Kansas, USA) and mounted a motorised treadmill (Pulsar 3 p, H/P/Cosmos, Nussdorf-Traunstein, Germany) in temperate conditions (20°C–22°C, 50% RH). $\dot{V}O_{2peak}$ was then assessed using a metabolic cart (TrueOne, Parvomedics, Sandy, Utah, USA). The incremental exercise test

REG	**45 min**			**15 min**	**45 min**		
$BRK_{no\text{-}cool}$	**30 min**	**3 min**	**15 min**	**15 min**	**30 min**	**3 min**	**15 min**
BRK_{cool}	**30 min**	**3 min**	**15 min**	**15 min**	**30 min**	**3 min**	**15 min**
$ExtHT_{only}$	**45 min**			**20 min**	**45 min**		
$ExtHT_{cool}$	**30 min**	**3 min**	**15 min**	**20 min**	**30 min**	**3 min**	**15 min**

Figure 1 Schematic overview of the five experimental trials. In the trials with 3 min breaks in the environmental chamber (40°C, 41% RH), blue represents cooling where cold fluid (5 mL/kg of 13°C water) and an ice-towel were provided, and red represents breaks without fluid or an ice-towel provided. At each half-time break (green), participants moved to temperate conditions (~22°C) and consumed 8 mL/kg of 13°C water. REG regular match; $BRK_{no\text{-}cool}$ 3 min breaks no cooling; BRK_{cool} 3 min cooling breaks; $ExtHT_{only}$ extended half-time; $ExtHT_{cool}$ 3 min cooling breaks and extended half-time. BRK_{cool}, breaks with cooling; $BRK_{no\text{-}cool}$, breaks without cooling; $ExtHT_{cool}$, cooling breaks with 5 min extended half-time; $ExtHT_{only}$, extended half-time without cooling breaks; RH, relative humidity.

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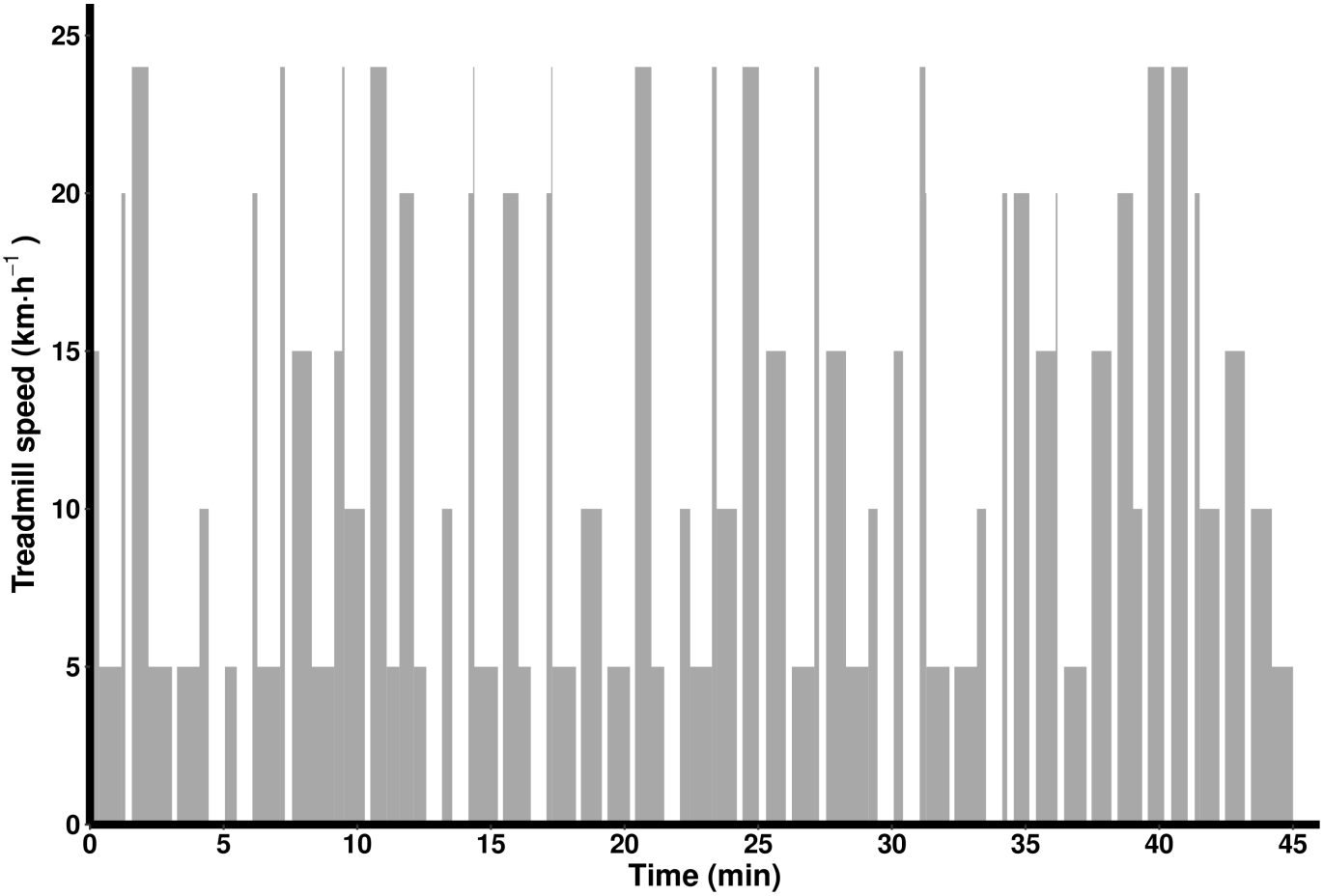

Figure 2 Schematic overview of the intermittent treadmill protocol for the regular trial. Both halves were identical. All trials followed the same intermittent protocol, with three trials having 3 min breaks 30 min into each half. The designated speeds for each segment of the protocol, the time spent at each speed when attained, the number of times they featured in each half, and the percentage of total time spent at each designated speed were: 0 km/hour (17 s, n=32, 26%), 5 km/hour (40 s, n=20, 33%), 10 km/hour (30 s, n=11, 15%), 15 km/hour (22 s, n=9, 11%), 20 km/hour (4 s, n=12, 7%) and 24 km/hour (3 s, n=13, 8%). Treadmill acceleration and deceleration were set to 0.34 m·s^2 (level 3) and −0.69 m·s^2 (level 4), respectively, and included in the width of each bar and the percentages.

was performed at a predetermined self-selected speed with the gradient starting at 0% and increasing by 1% every 1 min until volitional exhaustion.

Experimental trials

On arrival at the laboratory, participants provided a urine sample to assess urine specific gravity (USG: PEN-Urine S.G.; Atago Co, Tokyo, Japan). If USG was >1.025, participants consumed 5 mL/kg of water during instrumentation, ~30 min before exercise commenced. Participants then self-inserted a telemetric capsule (e-Celsius Performance; BodyCap, Caen, France) the length of one index finger beyond the anal sphincter to determine T_{re}, which was logged every 30 s. Participants were then provided with a heart rate monitor chest strap and a preweighed football ensemble, including a standardised jersey, shorts, and socks (100% polyester, adidas, Herzogenaurach, Germany). Participants wore the same pair of their own running shoes for each test. Following the recording of body mass, participants were fitted with four T_{sk} sensors (iButtons, Maximum Integrated Products, San Jose, California, USA) affixed using hypoallergenic tape on the chest, arm, mid-thigh and calf to determine mean T_{sk}.[23] Thereafter, participants lay supine for 5 min to allow for baseline measures to be recorded, before entering the environmental chamber and immediately beginning a 10 min standardised warm-up. All warm-ups consisted of 5 min of running on the treadmill at 10 km/hour, then four 10 s accelerations (2×15 km/hour, 1×17 km/hour, 1×20 km/hour), each followed by 50 s rest. During each trial, rating of perceived exertion (RPE),[24] thermal comfort[25] and ambient conditions (ie, temperature and RH) were recorded every 10 min. T_{sk} and heart rate were logged continuously and recorded as 1 min and 5 min averages, respectively. T_{re} was continuously monitored during each protocol and participants were removed from the environmental chamber if T_{re} reached the cut-off of 40.0°C (n=1). On completion of the intermittent protocol, participants exited the environmental chamber and were weighed to assess changes in body mass. Whole-body sweat rate (WBSR) was calculated accounting for the water ingested, sweat trapped in the football ensemble and time spent exercising.[26]

The regular match simulation (REG; figure 2) included no breaks except for the 15 min half-time, where 13 min was spent seated outside the environmental chamber in an air-conditioned room (~22°C), with the first and last min spent leaving and entering the environmental chamber. During half-time participants were provided with 8 mL/kg of 13°C water. The $BRK_{no\text{-}cool}$ trial included 3 min breaks in each half, where participants got off the treadmill to avoid direct airflow but were not provided with cold fluid or an ice towel during the breaks. The half-time

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break was 15 min. During the break with cooling (BRK_{cool}) trial, participants were provided with 5 mL/kg of 13°C water and an ice towel, which was draped over their shoulders during the 3 min breaks. The half-time break was 15 min. The ice towel (100% cotton) was soaked in cold water, wrung and evenly filled with ~2.5 kg of crushed ice.[9 27] Ice towels were donned for 2.5 min during the 3 min break. The extended half-time without additional cooling breaks ($ExtHT_{only}$) trial featured no 3 min breaks, but half-time was extended by 5 min (20 min total half-time). The extended half-time with cooling breaks ($ExtHT_{cool}$) trial included both the 3 min BRK_{cool} and the extended half-time.

Equity, diversity and inclusion

Our study included young males with a second study conducted in young females. The two groups were separated due to the different fitness level (ie, $\dot{V}O_{2peak}$)[28 29] and physical demands[21 30 31] of elite male and female football matches. The research team consisted of a mixture of early career (two) and mid-career (two) researchers, established academics (ie, professors; three) and an industry-based practitioner of different nationalities. Author disciplines include thermal/environmental and exercise physiologists and sports medicine physicians.

Patient and public involvement

This research was completed without patient or public involvement.

Statistical analyses

An a priori power analysis for sample size estimation was conducted for the difference in core temperature rise from the 30th to the 90th min between a regular simulated football match and one that included cooling breaks, which provided a Hedges g of 0.84.[9] With an alpha of 0.05 and power of 0.80, the required sample size needed for repeated measures within group comparisons was 10 (G Power software V.3.1.9.7). All statistical analyses were performed in R statistical software (V.4.2.1).[32] Linear mixed effect models were estimated using the 'lme4' package,[33] using restricted maximum likelihood and 'nloptwrap' optimiser, with each model including participant ID as a random intercept. Q-Q plots and histograms were used to assess data normality. T_{re}, T_{sk} and heart rate were analysed using linear mixed-effects models with time (5 min increments) and protocol type as fixed effects. Time was implemented as an ordered factor. Thermal comfort and RPE were analysed similarly but with time in 10 min increments. Pretrial USG, resting T_{re} and WBSR were analysed using a linear mixed effect model with protocol type as a fixed effect. Pairwise comparisons were conducted using the 'emmeans' package[34] and the Kenward-Roger method was used to approximate df. Significance was accepted at $p \leq 0.05$ for the α-error. Data are presented as means and 95% CIs, unless otherwise indicated. Relationships between selected physiological variables were evaluated using Pearson's correlation coefficient (r). Data were visualised using the 'ggplot2' package.[35] Statistical analyses and the presentation of results are consistent with the checklist for statistical assessment of medical papers statement[36] and can be found in online supplemental file 1.

RESULTS

Participants presented with a similar pretrial USG (mean of trials: 1.015±0.009; all p≥0.053) and resting T_{re} (mean of trials: 37.1°C±0.3°C; all p≥0.158). The percentage of maximum heart rate in REG was 76%±7% in the first half and 78%±7% in the second half. During two of the $BRK_{no\text{-}cool}$ trials, two participants did not complete the protocol. One participant was removed from the environmental chamber at 80 min on reaching a T_{re} of 40.0°C. The other was removed after 60 min due to lightheadedness (final T_{re}=38.8°C). Data from both participants were included in all analyses.

Mean T_{re}, T_{sk} and heart rate for each trial are presented in table 1, with values across the 90 min simulations presented in figures 3 and 4. Final T_{re} was 39.1°C (38.8, 39.3) in REG, 38.9°C (38.6, 39.1) in $BRK_{no\text{-}cool}$, 38.7°C (38.4, 38.9) in BRK_{cool}, 39.0°C (38.7, 39.2) in $ExtHT_{only}$ and 38.7°C (38.4, 38.9) in $ExtHT_{cool}$ (figure 3A). Final T_{re} was significantly lower in $BRK_{no\text{-}cool}$ (−0.20°C (−0.39 to −0.01), p=0.038), BRK_{cool} (−0.39°C (−0.57 to −0.21), p<0.001) and $ExtHT_{cool}$ (−0.40°C (−0.58 to −0.22), p<0.001) than REG, but not in $ExtHT_{only}$ (p=0.383; figure 3A). Final T_{re} in BRK_{cool} was 0.19°C (−0.00, 0.38) lower than $BRK_{no\text{-}cool}$ (p=0.056). Final heart was significantly lower in BRK_{cool} (−10 beats/min (−16, −4), p=0.001) and $ExtHT_{cool}$ (−10 beats/min (−16, −4), p<0.001) than REG (171 beats/min (162, 180)), but not in $BRK_{no\text{-}cool}$ and $ExtHT_{only}$ (p≥0.19; figure 4).

WBSR rate was similar across all trials (1.7±0.5 L/hour; all p≥0.07). The amount of fluid ingested was significantly higher in BRK_{cool} (1241 mL (1143, 1338)) and $ExtHT_{cool}$ (1229 mL (1132, 1327)) compared with REG (595 mL (497, 692)), $BRK_{no\text{-}cool}$ (584 mL (486, 681)) and $ExtHT_{only}$ (592 mL (495, 690); all p<0.001), but there were no significant differences between BRK_{cool} and $ExtHT_{cool}$ (p=0.773) and REG, $BRK_{no\text{-}cool}$, and $ExtHT_{only}$ (p≥0.781). Body mass loss in BRK_{cool} (−1.7% (−2.1, −1.3)) and $ExtHT_{cool}$ (−1.9% (−2.3, −1.5)) was significantly lower than in REG (−2.7% (−3.1, −2.3)), $BRK_{no\text{-}cool}$ (−2.5% (−2.9, −2.1)) and $ExtHT_{only}$ (−2.5% (−2.9, −2.1); all p≤0.001). Per cent body mass loss was correlated with mean heart rate (r=−0.55, p<0.001).

Mean RPE was significantly lower in BRK_{cool} (−0.4 (−0.7, −0.1), p=0.004) and $ExtHT_{cool}$ (−0.5 (−0.7, −0.2), p=0.002) than REG (12.9 (12.1, 13.6)), with $ExtHT_{cool}$ also being lower than $ExtHT_{only}$ (−0.3 (−0.6, −0.0), p=0.03). Mean thermal comfort was significantly lower in BRK_{cool} (5.1 (4.8, 5.5)) and $ExtHT_{cool}$ (5.1 (4.7, 5.4)) than REG (−0.3 (−0.4, −0.1) and −0.3 (−0.5 to −0.2), p≤0.001), $BRK_{no\text{-}cool}$ (−0.3 (−0.4, −0.2) and −0.4 (−0.5 to −0.2), p≤0.001) and $ExtHT_{only}$ (−0.2 (−0.3, −0.1) and −0.3 (−0.4 to −0.2), p≤0.001; table 2).

Table 1 Mean rectal temperature, skin temperature and heart rate during five simulated football matches at 40°C and 41% relative humidity with and without cooling breaks and an extended half-time

Outcome variable	REG	$BRK_{no\text{-}cool}$	BRK_{cool}	$ExtHT_{only}$	$ExtHT_{cool}$	Trial comparisons
Mean T_{re} (°C)	38.5 (38.3, 38.7)	38.4 (38.2, 38.6)	38.3 (38.1, 38.5)	38.4 (38.2, 38.6)	38.2 (38.0, 38.4)	A>B, D>C>E
Mean T_{sk} (°C)	35.5 (35.1, 35.9)	35.6 (35.2, 36.0)	35.1 (34.7, 35.6)	35.4 (35.0, 35.8)	34.9 (34.5, 35.3)	B>A > D>C>E
Mean HR (beats·min^{-1})	147 (139, 155)	145 (137, 153)	142 (133, 150)	147 (139, 155)	140 (132, 148)	A, D>B >C>E

Data are presented as mean and 95% CI. Significance is denoted as '>' and was set at p≤0.05. A: regular match (REG), B: 3 min breaks no cooling ($BRK_{no\text{-}cool}$), C: 3 min cooling breaks (BRK_{cool}), D: extended half-time ($ExtHT_{only}$), E: 3 min cooling breaks and extended half-time ($ExtHT_{cool}$).

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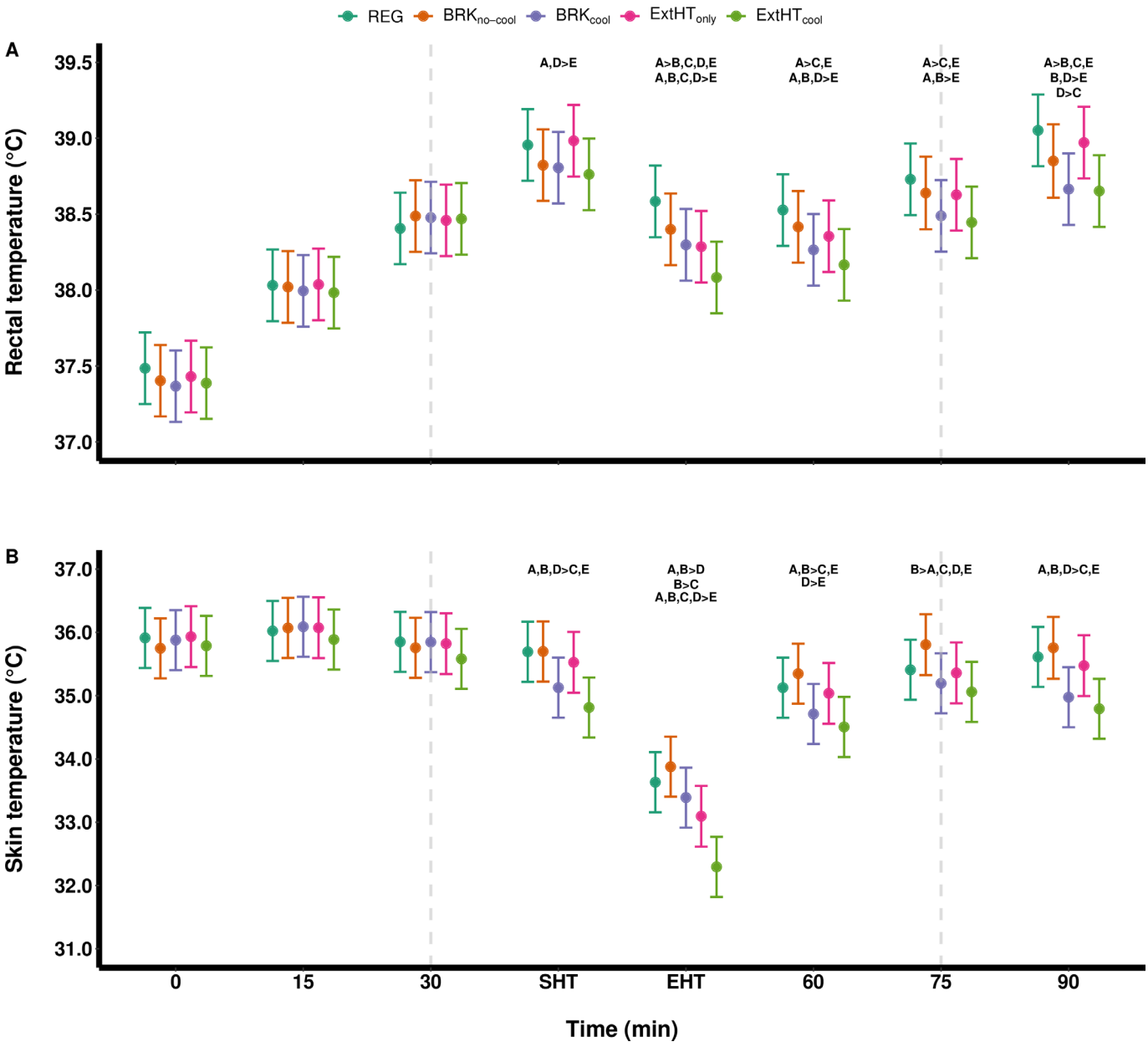

Figure 3 Rectal (A) and skin (B) temperature during five simulated football matches at 40°C and 41% relative humidity with and without cooling breaks and an extended half-time. Data are presented as mean and 95% CIs. Hashed lines represent cooling break implementation, where applicable. Significance is denoted as '>' and was set at p≤0.05. (A) Regular match (REG), (B) 3 min breaks no cooling ($BRK_{no-cool}$), (C) 3 min cooling breaks (BRK_{cool}), (D) extended half-time ($ExtHT_{only}$), (E) 3 min cooling breaks and extended half-time ($ExtHT_{cool}$). EHT, end of half time; BRK_{cool}, breaks with cooling; $BRK_{no-cool}$, breaks without cooling; $ExtHT_{cool}$, cooling breaks with 5 min extended half-time; $ExtHT_{only}$, extended half-time without cooling breaks; SHT, start of half time.

DISCUSSION

This study evaluated the efficacy of the FIFA cooling break heat policy in attenuating thermal strain during an intermittent treadmill protocol in 32°C WBGT conditions. In comparison to a regular match (REG), implementing 3 min cooling breaks (BRK_{cool}) and combining the breaks with a 5 min longer half-time ($ExtHT_{cool}$) decreased final T_{re} by 0.4°C. Implementation of both cooling strategies either individually or in combination also lowered mean T_{re} (~0.2°C), T_{sk} (~0.5°C), heart rate (~6 beats/min), RPE and improved thermal comfort, relative to REG. Although $BRK_{no-cool}$ and $ExtHT_{only}$ lowered mean T_{re} (~0.09°C) and $BRK_{no-cool}$ reduced heart rate (3 beats/min) relative to REG, these adjustments were less pronounced. Hence, the current FIFA heat policy (ie, BRK_{cool}) appears to provide a low resource strategy to attenuate the development of thermal, cardiovascular and perceptual (ie, RPE and thermal comfort) strain during a simulated football match in the heat, with a 5 min half-time extension further attenuating thermal (T_{re}: ~0.05°C, T_{sk}: ~0.3°C) and cardiovascular strain (heart rate: ~2 beats/min). However, while the implementation of BRK_{cool} and $ExtHT_{cool}$ had a cooling effect, additional and/or more aggressive cooling strategies may be required in field settings or during play under more severe environmental heat stress, which would elicit greater thermal strain.

The peak T_{re} reached in REG (39.1°C) is comparable to that of a previous outdoor match conducted in a 34°C environment

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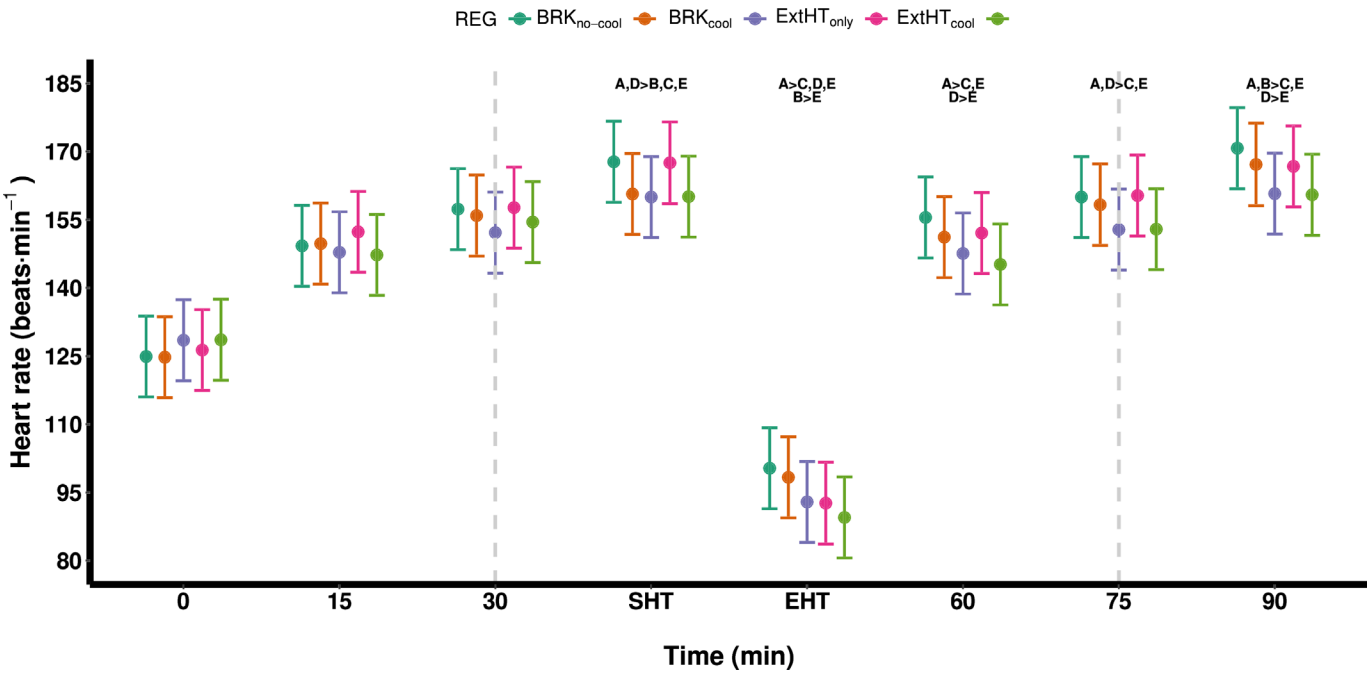

Figure 4 Heart rate during five simulated football matches at 40°C and 41% relative humidity with and without cooling breaks and an extended half-time. Data are presented as mean and 95% CI. Hashed lines represent cooling break implementation, where applicable. Significance is denoted as '>' and was set at p≤0.05. (A) Regular match (REG), (B) 3 min breaks no cooling ($BRK_{no\text{-}cool}$), (C) 3 min cooling breaks (BRK_{cool}), (D) extended half-time ($ExtHT_{only}$), (E) 3 min cooling breaks and extended half-time ($ExtHT_{cool}$). EHT, end of half time; BRK_{cool}, breaks with cooling; $BRK_{no\text{-}cool}$, breaks without cooling; $ExtHT_{cool}$, cooling breaks with 5n min extended half-time; $ExtHT_{only}$, extended half-time without cooling breaks; SHT, start of half time.

(~27°C WBGT),[6] but lower than that attained in hotter environments (36°C–43°C; 29°C–32°C WBGT) where peak rectal and gastrointestinal temperature exceeded 39.5°C.[5 6] This discrepancy in peak core temperature might indicate an effect of radiative heat load from direct exposure to the sun during outdoor play and/or differences in air velocity (ie, wind speed), which requires further examination. Regardless, final T_{re} in $BRK_{no\text{-}cool}$ (~0.20°C), BRK_{cool} (~0.39°C) and $ExtHT_{cool}$ (~0.40°C) were lower than in REG, with $BRK_{no\text{-}cool}$ being significantly higher than $ExtHT_{cool}$ but not BRK_{cool} (p=0.056). Despite the ~0.19°C difference in final T_{re} between BRK_{cool} and $BRK_{no\text{-}cool}$ not reaching statistical significance, the ingestion of cold water and an ice towel draped over the shoulders appears to provide greater cooling power during the 3 min breaks. Our data also indicated that compared with REG, BRK_{cool} lowered mean T_{re} by ~0.17°C and $ExtHT_{cool}$ by ~0.22°C, whereas $BRK_{no\text{-}cool}$ (0.09°C) and $ExtHT_{only}$ (0.08°C) had a minimal effect (table 1 and figure 3A). Notably, both BRK_{cool} and $ExtHT_{cool}$ decreased T_{sk} by ~0.5°C (figure 3B). The reduction in whole-body temperature (ie, T_{re} and T_{sk}) provided by BRK_{cool} and $ExtHT_{cool}$ is of clinical relevance and likely to decrease the risk of exertional heat illness. This reduced risk is further evidenced by a lower cardiovascular and perceptual strain (tables 1 and 2). Previously, Chalmers *et al*[9] demonstrated that compared with a regular match, both an $ExtHT_{only}$ and 3 min cooling breaks lowered T_{re} from the 30th to the 90th min by ~0.21°C and ~0.28°C, respectively. Our findings partly align with this observation, as $ExtHT_{only}$ attenuated the rise in T_{re} from the 30th to the 90th min by ~0.13°C, compared with REG. However, BRK_{cool} in the current study provided a larger cooling effect, attenuating the rise in T_{re} from the 30th to the 90th min by ~0.46°C. The greater attenuation, compared with Chalmers *et al*,[9] may relate to the higher absolute T_{re} in the 30th min noted in the current study (~38.5°C vs ~38.0°C), which likely resulted from the hotter environment (32°C vs 30°C WBGT) and higher initial T_{re} due to the inclusion of a warm-up. The combination of these factors may have provided the potential for a larger ergogenic benefit of the cooling break.[37] Additionally, each 3 min cooling break in the current study provided ~135 mL more 13°C water to be consumed, contributing to the overall cooling effect. Interestingly, however, $ExtHT_{cool}$ did not further mitigate the rise in T_{re} from the 30th to the 90th min, compared with BRK_{cool}. As such, implementing a cooling strategy during the extended half-time may attenuate thermal strain to a greater extent than BRK_{cool} or $ExtHT_{cool}$.

Heart rate was lower during BRK_{cool} (5 beats/min) and $ExtHT_{cool}$ (7 beats/min) compared with REG (figure 4), which is slightly less than noted by Chalmers *et al*[9] when implementing 3 min cooling breaks (−10 beats/min) and $ExtHT_{only}$ (−8 beats/min),

Table 2 Mean rating of perceived exertion and thermal comfort during five simulated football matches at 40°C and 41% relative humidity with and without cooling breaks and an extended half-time

Outcome variable	REG	$BRK_{no\text{-}cool}$	BRK_{cool}	$ExtHT_{only}$	$ExtHT_{cool}$	Trial comparisons
Rating of perceived exertion	12.9 (12.1, 13.6)	12.7 (11.9, 13.5)	12.5 (11.7, 13.2)	12.7 (12.0, 13.5)	12.4 (11.7, 13.2)	A>C, E D>E
Thermal comfort	5.4 (5.1, 5.8)	5.4 (5.1, 5.8)	5.1 (4.8, 5.5)	5.4 (5.0, 5.7)	5.1 (4.7, 5.4)	A, B, D>C, E

Data are presented as mean and 95% CI. Significance is denoted as '>' and was set at p≤0.05. A: regular match (REG), B: 3 min breaks no cooling ($BRK_{no\text{-}cool}$), C: 3 min cooling breaks (BRK_{cool}), D: extended half-time ($ExtHT_{only}$), E: 3 min cooling breaks and extended half-time ($ExtHT_{cool}$).

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relative to a regular match. RPE and thermal comfort were lower only during BRK_{cool} and $ExtHT_{cool}$, compared with REG (table 2). However, the addition of an extended half-time alongside 3 min cooling breaks did not further improve perceptual responses than the cooling breaks alone. An ice towel donned around the neck and shoulders has previously been shown to reduce thermal sensation during simulated rugby league matches,[38] but not during simulated football matches.[9] The reduction in thermal sensation and thermal comfort are often associated with reductions in T_{sk}, especially when cooling is applied to the neck and chest regions where there is a higher density of cold thermoreceptors.[39] In contrast, $BRK_{no\text{-}cool}$ and $ExtHT_{only}$ had no meaningful effect on T_{sk} (figure 3B), resulting in no improvements in perceptual strain compared with REG (table 2). During tennis matches completed in the heat (~34°C WBGT), players adapt their behaviour (ie, increase time between points) to minimise the adverse effects of the environment.[40] Similarly, elite male football players have been shown to run ~7% less during matches at 42°C, despite the quantity and length of sprints being similar to that of a match in 21°C,[5] whereas, high compared with moderate WBGT led to ~10% less sprints being completed in the 2014 Men's World Cup.[1] Given that the external workload was fixed across the trials in the current study, participants could not self-regulate their efforts. As such, the reductions in physiological and perceptual strain observed in both BRK_{cool} and $ExtHT_{cool}$ suggest that these cooling interventions could partly offset heat-related decrements in physical performance,[5 41] although further research is required to elucidate this effect in hot outdoor settings.

In the current study, WBSR was consistent across all trials. However, in BRK_{cool} and $ExtHT_{cool}$ fluid intake was significantly higher (~645 mL) than in the other trials. The additional fluid consumed in BRK_{cool} and $ExtHT_{cool}$ resulted in a lower per cent body mass loss of ~1.7% and ~1.9%, respectively, compared with the 2.5%–2.7% loss in $BRK_{no\text{-}cool}$, $ExtHT_{only}$ and REG. Per cent body mass loss was correlated with mean heart rate ($r=-0.55$, $p<0.001$), which suggests that a better hydration status, in conjunction with additional breaks in play, contributed to attenuate cardiovascular strain (figure 4). Previous studies have shown that a body mass loss of ~2.4% during football play impairs subsequent performance during skills tests by ~5%[42] and a Yo-Yo Intermittent Recovery Test by 13%,[43] with the former associated with a higher mean heart rate (~7 beats/min).[42] Of note, fluid intake was prescribed based on body mass during all experimental trials in the current study, which contrasts to the ad libitum nature of fluid consumption adopted during football matches.[44] Notwithstanding, the 1.2 L consumed in BRK_{cool} and $ExtHT_{cool}$ broadly aligns with the fluid consumption (0.9–2.6 L) reported during football matches in the heat, where ad libitum drinking was permitted.[5 44 45]

Limitations and practical implications

The main limitation of the study is that the trials were conducted indoors on a unidirectional, motorised treadmill, which is not akin to a competitive outdoor match-play against an opponent and potentially led to a lower peak T_{re}. However, the controlled laboratory setting, including a standardised warm-up and fluid provision, allowed for a systematic evaluation of the FIFA cooling break policy to that of a regular match and alternative approaches at a work rate and in environmental conditions comparable to outdoor play. Moreover, the half-time breaks were completed in an air-conditioned environment to align with what elite football players experience during matches. Although $ExtHT_{cool}$ provided additional benefits to BRK_{cool}, $ExtHT_{only}$ did not, such that it is not recommended to extend half-time instead of providing 3 min cooling breaks. Future research may seek to evaluate different cooling break approaches in real-world settings with elite-level football players,[10] although with caution as positional demands and tactical strategies during outdoor matches influence activity patterns (ie, metabolic heat production) and as such, physiological strain. Environmental conditions between outdoor matches may also differ and influence the development of physiological strain. Additional studies may also examine the effects of heat acclimation and acclimatisation on thermal and cardiovascular responses during football match-play in the heat and the interaction with cooling breaks to attenuate physiological strain. Finally, FIFA currently uses a WBGT threshold to determine when to implement the cooling break policy. However, WBGT only provides a measure of environmental heat stress and does not account for the interaction of the players within that environment (ie, metabolic heat production and clothing), and as such cannot adequately represent human heat strain.[46] The development of football-specific heat stress policy with actionable categories (eg, cooling break, cooling break with extended half-time and cooling, postponement/cancellation) that account for the evaporative heat loss requirement of players and the evaporative capacity of the environment, based on the prevailing ambient conditions and player activity and clothing, is warranted to safeguard their health.

CONCLUSION

Our findings indicate that in a simulated football match the current FIFA cooling break heat policy (BRK_{cool}) lowered final T_{re} (0.39°C), mean T_{re} (0.17°C), heart rate (five beats/min) and T_{sk} (0.4°C) and attenuated perceptual strain (RPE: 0.4, thermal comfort: 0.3) compared with a regular match conducted in the heat. The addition of a 5 min extension to half-time further lowered mean T_{re} (0.05°C), heart rate (2 beats/min) and T_{sk} (0.3°C). Although BRK_{cool} and $ExtHT_{cool}$ had a significant cooling effect during a football simulation in 32°C WBGT, additional and/or more aggressive cooling strategies may be required in more stressful environmental conditions, especially in field settings where a validation of the cooling breaks is required, as thermal strain appears to be greater.

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